

DECENT COACHING CLASSES

VIII CBSE

Mind Your Math - II

Subject: Mathematics — Squares and Square Roots

Time: 1 Hour

Total Marks: 30

Note: (i) All questions are compulsory. (ii) Section A carries 1 mark each, Section B carries 2 marks each, Section C carries 3 marks each and Section D carries 6 marks. (iii) Show all steps of working clearly.

Section A — Choose the correct alternative. [5]

- Which of the following cannot be a perfect square? (1)
(a) 1024 (b) 3843 (c) 7921 (d) 4096
- If a number has 3 or 7 in its unit's place, then its square ends in (1)
(a) 1 (b) 9 (c) 4 (d) 6
- The square root of 1024 by the prime factorisation method is (1)
(a) 30 (b) 32 (c) 34 (d) 36
- How many natural numbers lie between 11^2 and 12^2 ? (1)
(a) 20 (b) 21 (c) 22 (d) 23
- $\sqrt{0.0064} =$ (1)
(a) 0.8 (b) 0.08 (c) 0.008 (d) 8

Section B — Answer the following questions. [10]

- Without actual multiplication, find the square of 87 using the identity $(a + b)^2 = a^2 + 2ab + b^2$. (2)
- Is 2450 a perfect square? If not, find the smallest number by which it should be multiplied to get a perfect square. (2)
- Find the square root of 7056 by the division method. (2)
- The area of a square field is 5041 m^2 . Find the length of its side. (2)
- Find a Pythagorean triplet whose smallest member is 14. (2)

Section C — Answer the following questions. [9]

- By the long division method, find the square root of 15129. Show all the steps. (3)
- Find the smallest number by which 1152 must be divided so that the quotient is a perfect square. Also find the square root of the quotient. (3)
- Find the least number which must be added to 6412 to make it a perfect square. (3)

Section D — Answer the following question. [6]

- Answer the following:
(a) Find the square root of 2304 by the prime factorisation method. (3)
(b) A gardener has 1936 plants. He wants to plant them in such a way that the number of rows is equal to the number of columns. Find the number of rows. (3)

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Section A Answers

1. (b) 3843 2. (b) 9 3. (b) 32 4. (c) 22 5. (b) 0.08

Section B Answers

1. $87 = 80 + 7$. $(80+7)^2 = 80^2 + 2 \times 80 \times 7 + 7^2 = 6400 + 1120 + 49 = 7569$
2. $2450 = 2 \times 5^2 \times 7^2$. Since 2 has no pair, 2450 is not a perfect square. Multiplying by 2 gives $4900 = 2^2 \times 5^2 \times 7^2 = 70^2$, a perfect square. $\therefore$ Smallest number = 2
3. By division method, $\sqrt{7056} = 84$ (Check: $84^2 = 7056$)
4. Side = $\sqrt{5041} = 71$ m (Check: $71^2 = 5041$)
5. Using $2m$, m^2-1 , m^2+1 with $2m = 14 \Rightarrow m = 7$. $m^2-1 = 48$, $m^2+1 = 50$. $\therefore$ Triplet is 14, 48, 50 (Check: $14^2+48^2 = 196+2304 = 2500 = 50^2$)

Section C Answers

1. By long division method, $\sqrt{15129} = 123$ (Check: $123^2 = 15129$)
2. $1152 = 2^7 \times 3^2$. The power of 2 is odd, so dividing by 2 gives $576 = 2^6 \times 3^2$, a perfect square. $\therefore$ Smallest number = 2, and $\sqrt{576} = 24$
3. $\sqrt{6412}$ lies between 80 ($80^2 = 6400$) and 81 ($81^2 = 6561$). The next perfect square after 6412 is 6561. $\therefore$ Number to be added = $6561 - 6412 = 149$

Section D Answer

- (a) $2304 = 2^8 \times 3^2$ (by prime factorisation). $\sqrt{2304} = 2^4 \times 3 = 16 \times 3 = 48$ (Check: $48^2 = 2304$)
- (b) Since rows = columns, the total number of plants is a perfect square, and the number of rows = $\sqrt{1936} = 44$ (Check: $44^2 = 1936$)